Technical Design Document (TDD)

1. Project identification

Item	Information
Project title	KOHA-Lite Inventory Management for Little Free Libraries
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Course/section	01:198:437:01
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Version	1.0

2. Design overview

KOHA-Lite is a relational database for recorded circulation across participating Little Free Libraries. It stores users and roles, director-library assignments, libraries, book titles, individual copies, loans, donations, explicit visits, inventory corrections, and audit events. MySQL 8.0/InnoDB provides primary and foreign keys, unique and check constraints, generated columns, views, stored routines, triggers, transactions, locking reads, roles, and grants.

The implementation contains fourteen base tables. Availability and quantity are derived from physical-copy rows. State-changing workflows use stored procedures so authorization, validation, the business change, and its audit record occur through one controlled operation. Checkout locks the selected copy using `SELECT ... FOR UPDATE`; a generated unique key independently limits each copy to one active loan.

3. Technical environment

Component	Selected technology/tool	Purpose
DBMS	MySQL Server 8.0+ / InnoDB	Relational storage, constraints, transactions, routines, roles
SQL client	MySQL command-line client; MySQL Workbench optional	Build, query, and inspect the database
Modeling	draw.io-compatible XML; Mermaid review source	Editable ERD and portable diagram source
Documentation	Markdown and PDF	Versionable sources and submission artifacts
Build	PowerShell, Git, exact Git tag	Reproducible fresh database creation
Source control	Git/GitHub after repository initialization	Immutable release tag and history
Operating environment	Windows; MySQL 8.x	Primary classroom/test platform

4. Architecture and scope

The project uses one database-centered MySQL schema named `koha_lite`. A future application would call controlled procedures and read protected views. Individual patrons are represented by business records, not necessarily individual MySQL accounts. Representative MySQL roles demonstrate database permissions, while `UserRole` and `DirectorLibrary` enforce business identity and row scope.

Included artifacts are schema objects, routines, triggers, indexes, roles/grants, fictional sample data, reports, validation queries, smoke tests, documentation, ERD, and an exact-tag build script. A web/mobile UI, geographic map, production identity provider, notifications, surveillance, and production backup automation are excluded.

5. Entity-relationship design

The editable diagram is `docs/KOHA-Lite_ERD.drawio`; `docs/KOHA-Lite_ERD.mmd` is a reviewable equivalent.

Entity	Purpose	Primary key	Important relationships
UserAccount	Synthetic project user	UserID	Roles, director assignments, loans, donations, visits, adjustments, audit events
BusinessRole	Approved business role	RoleCode	Many-to-many with users through UserRole
UserRole	User/role assignment	UserID + RoleCode	Resolves UserAccount-to-BusinessRole many-to-many
Library	Participating physical location	LibraryID	Copies, circulation locations, visits, donations, assignments
DirectorLibrary	Director's managed library	UserID + LibraryID	Resolves director-to-library many-to-many
BookCategory	Controlled title classification	CategoryID	One category to many titles
BookTitle	Shared bibliographic description	BookTitleID	One title to many copies and donation items
BookCopy	Individually tracked physical copy	BookCopyID	One title; current library; loan and adjustment history
Loan	Checkout/return lifecycle	LoanID	One copy, one patron, checkout and optional return libraries/actors
Donation	Donation submission/decision	DonationID	One donor, one receiving library, many donation items
DonationItem	Donated title and quantities	DonationItemID	One donation, one title, zero or more accepted copies
Visit	Explicit patron visit	VisitID	One patron, library, and recording actor
InventoryAdjustment	Immutable reconciliation event	AdjustmentID	One copy and director; prior/new state
AuditEvent	Append-only business-event evidence	AuditEventID	One actor; optional library; polymorphic affected record identifier

Cardinality summary:

- One user has zero-to-many `UserRole` rows; one business role has zero-to-many user assignments.
- One director and one library each have zero-to-many `DirectorLibrary` assignments.
- One category has many titles; one title has many physical copies.
- One copy has many historical loans but at most one active loan.
- One patron has many loans and visits.
- One donation has one or more donation items; one accepted donation item can create many copies.
- One copy has many inventory adjustments and audit events over time.

6. Relational schema

```
BusinessRole(RoleCode PK, RoleName UQ, RoleDescription)
UserAccount(UserID PK, DisplayName, Email UQ, IsActive, CreatedAt)
UserRole(UserID PK/FK -> UserAccount, RoleCode PK/FK -> BusinessRole, AssignedAt)
Library(LibraryID PK, LibraryName UQ, AddressLine1, City, StateCode, PostalCode, LibraryStatus, CreatedAt)
DirectorLibrary(UserID PK/FK -> UserAccount, LibraryID PK/FK -> Library, AssignedAt)
BookCategory(CategoryID PK, CategoryName UQ, CategoryDescription)
BookTitle(BookTitleID PK, ISBN13 UQ, Title, AuthorName, CategoryID FK -> BookCategory, CreatedAt)
Donation(DonationID PK, DonorUserID FK, ReceivingLibraryID FK, DonationStatus,
         SubmittedAt, DecidedAt, DecidedByUserID FK, DecisionNote)
DonationItem(DonationItemID PK, DonationID FK, BookTitleID FK,
             OfferedQuantity, AcceptedQuantity, UQ(DonationID, BookTitleID))
BookCopy(BookCopyID PK, BookTitleID FK, CurrentLibraryID FK, CopyStatus,
         RegistrationSource, DonationItemID FK, RegisteredByUserID FK, RegisteredAt)
Loan(LoanID PK, BookCopyID FK, PatronUserID FK, CheckoutLibraryID FK,
     CheckoutRecordedByUserID FK, CheckoutAt, ReturnLibraryID FK,
     ReturnRecordedByUserID FK, ReturnedAt, LoanStatus,
```

```
     ActiveBookCopyID GENERATED UQ)
Visit(VisitID PK, PatronUserID FK, LibraryID FK, VisitedAt,
      RecordedByUserID FK, RecordingMethod)
InventoryAdjustment(AdjustmentID PK, BookCopyID FK, DirectorUserID FK,
                    PreviousLibraryID FK, NewLibraryID FK,
                    PreviousStatus, NewStatus, Reason, AdjustedAt)
AuditEvent(AuditEventID PK, ActorUserID FK, EventType, EntityType, EntityID,
           EventTimestamp, Outcome, LibraryID FK, ContextData JSON)
```

7. Table and attribute specifications

BusinessRole

One row represents an approved business role.

Column	Type	Null	Constraint/meaning
RoleCode	VARCHAR(20)	No	PK; stable code such as PATRON
RoleName	VARCHAR(60)	No	UNIQUE display name
RoleDescription	VARCHAR(255)	No	Role purpose

UserAccount

One row represents a fictional registered user.

Column	Type	Null	Constraint/meaning
UserID	BIGINT UNSIGNED	No	PK, auto-increment
DisplayName	VARCHAR(120)	No	Nonblank synthetic name
Email	VARCHAR(255)	No	UNIQUE synthetic email; simple format check
IsActive	BOOLEAN	No	Active users may operate; default true
CreatedAt	DATETIME(6)	No	Creation time; default current timestamp

UserRole

One row assigns one business role to one user.

Column	Type	Null	Constraint/meaning
UserID	BIGINT UNSIGNED	No	PK/FK to UserAccount
RoleCode	VARCHAR(20)	No	PK/FK to BusinessRole
AssignedAt	DATETIME(6)	No	Assignment time

Library

One row represents one participating New Jersey Little Free Library.

Column	Type	Null	Constraint/meaning
LibraryID	BIGINT UNSIGNED	No	PK, auto-increment
LibraryName	VARCHAR(150)	No	UNIQUE, nonblank name
AddressLine1	VARCHAR(150)	No	Fictional/demo address line
City	VARCHAR(100)	No	City
StateCode	CHAR(2)	No	CHECK NJ
PostalCode	VARCHAR(10)	No	Postal code

Column	Type	Null	Constraint/meaning
LibraryStatus	ENUM	No	ACTIVE or INACTIVE
CreatedAt	DATETIME(6)	No	Creation time

DirectorLibrary

One row authorizes one director for one library.

Column	Type	Null	Constraint/meaning
UserID	BIGINT UNSIGNED	No	PK/FK to UserAccount
LibraryID	BIGINT UNSIGNED	No	PK/FK to Library
AssignedAt	DATETIME(6)	No	Assignment time

BookCategory

Column	Type	Null	Constraint/meaning
CategoryID	SMALLINT UNSIGNED	No	PK, auto-increment
CategoryName	VARCHAR(80)	No	UNIQUE category name
CategoryDescription	VARCHAR(255)	Yes	Optional explanation

BookTitle

One row stores bibliographic information shared by multiple copies.

Column	Type	Null	Constraint/meaning
BookTitleID	BIGINT UNSIGNED	No	PK, auto-increment
ISBN13	CHAR(13)	Yes	UNIQUE when supplied; 13-digit check
Title	VARCHAR(255)	No	Nonblank title
AuthorName	VARCHAR(180)	No	Nonblank author/display attribution
CategoryID	SMALLINT UNSIGNED	No	FK to BookCategory
CreatedAt	DATETIME(6)	No	Creation time

Donation

One row represents a submission and its final decision.

Column	Type	Null	Constraint/meaning
DonationID	BIGINT UNSIGNED	No	PK, auto-increment
DonorUserID	BIGINT UNSIGNED	No	FK to UserAccount
ReceivingLibraryID	BIGINT UNSIGNED	No	FK to Library
DonationStatus	ENUM	No	SUBMITTED, ACCEPTED, or REJECTED
SubmittedAt	DATETIME(6)	No	Submission time
DecidedAt	DATETIME(6)	Yes	Required after decision
DecidedByUserID	BIGINT UNSIGNED	Yes	FK; required after decision
DecisionNote	VARCHAR(500)	Yes	Acceptance/rejection context

DonationItem

One row represents one title offered within a donation.

Column	Type	Null	Constraint/meaning
DonationItemID	BIGINT UNSIGNED	No	PK, auto-increment
DonationID	BIGINT UNSIGNED	No	FK to Donation
BookTitleID	BIGINT UNSIGNED	No	FK to BookTitle
OfferedQuantity	SMALLINT UNSIGNED	No	CHECK greater than zero
AcceptedQuantity	SMALLINT UNSIGNED	Yes	Null before decision; not greater than offered

`UNIQUE(DonationID, BookTitleID)` prevents duplicate title lines.

BookCopy

One row represents one physical copy.

Column	Type	Null	Constraint/meaning
BookCopyID	BIGINT UNSIGNED	No	PK, auto-increment
BookTitleID	BIGINT UNSIGNED	No	FK to BookTitle
CurrentLibraryID	BIGINT UNSIGNED	Yes	FK to recorded location; required for available/on-loan copies
CopyStatus	ENUM	No	AVAILABLE, ON_LOAN, LOST, or RETIRED
RegistrationSource	ENUM	No	DIRECT or DONATION
DonationItemID	BIGINT UNSIGNED	Yes	FK required only for donation source
RegisteredByUserID	BIGINT UNSIGNED	No	FK to registering director
RegisteredAt	DATETIME(6)	No	Registration time

Loan

One row represents one checkout and its eventual return.

Column	Type	Null	Constraint/meaning
LoanID	BIGINT UNSIGNED	No	PK, auto-increment
BookCopyID	BIGINT UNSIGNED	No	FK to BookCopy
PatronUserID	BIGINT UNSIGNED	No	FK to borrowing patron
CheckoutLibraryID	BIGINT UNSIGNED	No	FK to originating library
CheckoutRecordedByUserID	BIGINT UNSIGNED	No	FK to patron/director actor
CheckoutAt	DATETIME(6)	No	Checkout timestamp
ReturnLibraryID	BIGINT UNSIGNED	Yes	FK; populated at return
ReturnRecordedByUserID	BIGINT UNSIGNED	Yes	FK; populated at return
ReturnedAt	DATETIME(6)	Yes	Return timestamp not before checkout
LoanStatus	ENUM	No	ACTIVE or RETURNED
ActiveBookCopyID	BIGINT UNSIGNED	Yes	Generated copy ID only for ACTIVE rows; UNIQUE

Visit

Column	Type	Null	Constraint/meaning
VisitID	BIGINT UNSIGNED	No	PK, auto-increment
PatronUserID	BIGINT UNSIGNED	No	FK to patron
LibraryID	BIGINT UNSIGNED	No	FK to visited library
VisitedAt	DATETIME(6)	No	Explicit visit time
RecordedByUserID	BIGINT UNSIGNED	No	FK to patron/director actor
RecordingMethod	ENUM	No	PATRON_SELF or DIRECTOR

InventoryAdjustment

One immutable row records a director's reconciliation.

Column	Type	Null	Constraint/meaning
AdjustmentID	BIGINT UNSIGNED	No	PK, auto-increment
BookCopyID	BIGINT UNSIGNED	No	FK to affected copy
DirectorUserID	BIGINT UNSIGNED	No	FK to actor
PreviousLibraryID	BIGINT UNSIGNED	Yes	FK to prior location
NewLibraryID	BIGINT UNSIGNED	Yes	FK to corrected location
PreviousStatus	ENUM	No	Prior copy status
NewStatus	ENUM	No	Corrected copy status
Reason	VARCHAR(500)	No	At least five non-space characters
AdjustedAt	DATETIME(6)	No	Adjustment time

AuditEvent

One immutable row records a business event.

Column	Type	Null	Constraint/meaning
AuditEventID	BIGINT UNSIGNED	No	PK, auto-increment
ActorUserID	BIGINT UNSIGNED	No	FK to actor
EventType	VARCHAR(60)	No	Stable nonblank event code
EntityType	VARCHAR(60)	No	Affected entity type
EntityID	BIGINT UNSIGNED	No	Affected record identifier
EventTimestamp	DATETIME(6)	No	Event time
Outcome	ENUM	No	SUCCEEDED or REJECTED; implemented business events use SUCCEEDED
LibraryID	BIGINT UNSIGNED	Yes	Optional FK for row-scope reporting
ContextData	JSON	Yes	Relevant IDs and prior/new context; no secrets

8. Normalization analysis

Normal form	Design check	Evidence
1NF	Atomic columns and no repeating groups	Roles, director assignments, donation items, copies, loans, and visits are separate rows rather than lists in User or Library.

Normal form	Design check	Evidence
2NF	Non-key values depend on the complete key	Composite-key tables UserRole and DirectorLibrary contain only facts about the complete assignment.
3NF	Non-key values do not depend on other non-key values	Category description is in BookCategory; bibliographic data is in BookTitle; copy state is in BookCopy; library names are not copied into Loan.

Stored copy quantity was deliberately rejected: counts are derived from `BookCopy`, preventing a stored quantity from disagreeing with individual copy state. `BookTitle` is separated from `BookCopy`, so three copies share one title without duplicating title/author/category. History tables retain foreign keys and event facts rather than mutable descriptive copies.

9. Integrity, validation, and business-rule enforcement

Rule	Technical implementation	Verification
BR-01 unique records/assignments	PKs, composite PKs, unique email/name/ISBN/title-line constraints	Duplicate inserts fail
BR-02 title and location integrity	NOT NULL title FK and copy-location CHECK	Invalid FK/location-state insert fails
BR-03 one active loan	Generated `ActiveBookCopyID` plus UNIQUE	Concurrent/duplicate checkout test
BR-04 available active-library checkout	`sp_CheckoutCopy`, role checks, locked copy read	Available succeeds; unavailable rejected
BR-05 atomic checkout	InnoDB transaction and EXIT handler rollback	Loan/copy/audit state checked together
BR-06 valid return	`sp_ReturnCopy` active-state and return-library validation	Valid cross-library return; duplicate rejected
BR-07 patron eligibility	`fn_HasRole` and active-user condition	Inactive/non-patron rejected
BR-08 scoped director	`fn_DirectorManagesLibrary` in director procedures	Unassigned director rejected
BR-09 donation quantities	CHECK constraints; acceptance copies offered quantity	Known donation fixture
BR-10 demand definition	COUNT(DISTINCT patron) with required half-open date range	Known report fixture
BR-11 inactive library	Procedure checks `LibraryStatus`	New activity rejected after deactivation
BR-12 attributable correction	`sp_AdjustInventory`, required reason, immutable row	Prior/new values and audit checked
BR-13 append-only audit	No DML grants; UPDATE/DELETE rejection triggers	Tamper test fails as expected
BR-14 preserve history	No cascade deletes; controlled state changes	Historical query remains available
BR-15 safe retries	State validation, unique active loan, rollback	Duplicate return/checkout changes nothing

10. Database objects and SQL design

Type	Object	Purpose	Related requirements
View	vw_AvailableCatalog	Available counts by library/title/category	FR-02, RQ-01
View	vw_ActiveLoan	Current loans and checkout context	FR-03, RQ-02
View	vw_LoanHistory	Complete circulation history	FR-07, RQ-03
View	vw_Inventory	Copy state, location, and active-loan flag	FR-09, RQ-06
View	vw_DonationStatus	Donation totals and state	FR-05, RQ-07
View	vw_AuditHistory	Human-readable append-only evidence	FR-11, RQ-08

Type	Object	Purpose	Related requirements
Function	fn_HasRole	Active-user business-role check	FR-03, FR-06, FR-10
Function	fn_DirectorManagesLibrary	Scoped director entitlement	FR-01, FR-05, FR-09, FR-12
Procedure	sp_RegisterBookCopy	Controlled direct copy registration	FR-01
Procedure	sp_CheckoutCopy	Atomic, concurrency-safe checkout	FR-03, FR-10, FR-13
Procedure	sp_ReturnCopy	Atomic cross-library return	FR-04, FR-10, FR-13
Procedure	sp_RecordVisit	Patron/director explicit check-in	FR-06
Procedures	sp_Submit/Accept/RejectDonation	Donation lifecycle and accepted copies	FR-05
Procedure	sp_AdjustInventory	Attributable reconciliation	FR-09
Procedure	sp_SetLibraryStatus	Scoped operational status change	FR-12
Triggers	AuditEvent/InventoryAdjustment no-update/no-delete	Append-only evidence	FR-09, FR-11
Indexes	Copy, loan, visit, donation, adjustment, audit indexes	Join/filter/order access paths	RQ-01 through RQ-08

The important checkout index path begins with copy identity for the locked point lookup. Patron history uses `(PatronUserID, CheckoutAt DESC)`, and available inventory uses `(CurrentLibraryID, CopyStatus, BookTitleID)`. Demand uses the checkout-library/time data; its performance should be evaluated with `EXPLAIN` as sample volume increases.

11. Query and report design

Query	Main tables	SQL features	Expected result
RQ-01 available catalog	Library, BookCopy, BookTitle, BookCategory	JOIN, WHERE, COUNT, GROUP BY, ORDER BY	Available title counts for one library/category
RQ-02 active loans	Loan, BookCopy, BookTitle, Library	JOIN, status filter, patron filter	Patron's active copies oldest first
RQ-03 borrowing history	Loan, BookCopy, BookTitle, two Library joins	JOIN, optional date range, ORDER BY	Checkout/return lifecycle
RQ-04 title demand	Loan, BookCopy, BookTitle	required date filter, optional library, COUNT DISTINCT, GROUP BY	Distinct patron and checkout counts
RQ-05 visit history	Visit, Library	patron/date filter, descending time	Explicit visits
RQ-06 inventory	BookCopy, BookTitle, Library, active Loan	joins, status/filter	Recorded copy state for assigned library
RQ-07 donations	Donation, DonationItem, BookTitle, Library	joins, aggregation, status filter	Offered/accepted quantities and state
RQ-08 audit	AuditEvent, UserAccount, Library	scoped filter, JSON context, time order	Attributable history

The validation script implements RQ-04 with a half-open interval (`>= start`, `< end`) to avoid overlapping date windows, and includes `EXPLAIN FORMAT=TREE` for patron loan history.

12. Security, privacy, backup, and recovery

- All sample names, addresses, and `.test` emails are fictional.
- Business authorization is enforced with `UserRole`, `DirectorLibrary`, functions, and stored procedures.
- Representative MySQL roles receive SELECT on protected views and EXECUTE on procedures; they receive no direct circulation-table modification rights.
- Audit and inventory-adjustment rows reject update and delete operations.

- Credentials are supplied via environment variables and never committed or written to logs.
- InnoDB transactions roll back incomplete multi-table workflows. The build stops on the first failed script.
- The build refuses to replace an existing database unless `ALLOW_DATABASE_RECREATE=YES` is explicitly set.
- Production encryption, identity federation, scheduled backup/restore testing, retention, and high availability remain outside this course implementation. A production deployment would require them.

13. Test plan and acceptance criteria

Test	Requirement/rule	Action	Expected result	Automated evidence
T-01	FR-01/BR-08	Assigned director registers copy	Copy and audit inserted	Procedure and fixture/query
T-02	FR-01/BR-08	Unassigned director registers copy	Rejected; no copy	Smoke test PASS
T-03	FR-02	Query known catalog fixture	Expected available counts	vw_AvailableCatalog
T-04	FR-03/BR-05	Checkout available copy	One loan; copy ON_LOAN; audit	Atomic-state PASS
T-05	FR-03/BR-03	Checkout already loaned copy	Rejected; one active loan	Failure test PASS
T-06	FR-04/BR-06	Return to different library	Loan returned; copy available there	Cross-library PASS
T-07	FR-13	Repeat completed return	Rejected; no extra change	Failure test PASS
T-08	FR-05/BR-09	Accept donation of two copies	Two linked copies; accepted count two	Donation PASS
T-09	FR-06	Self/director visit recording	Correct method, actor, patron, library	Seed/query evidence
T-10	FR-07/FR-10	Access another patron outside scope	Protected interface rejects/does not grant table DML	Role/grant inspection
T-11	FR-08/BR-10	Run Sept. fixture demand	Title 1 has two distinct patrons	Data-validation query
T-12	FR-09/BR-12	Adjust non-loaned copy with reason	Copy, adjustment, audit agree	Procedure/query
T-13	FR-11/BR-13	Update audit event	Trigger rejects update	Failure test PASS
T-14	FR-12/BR-11	Deactivate assigned library and submit activity	New activity rejected; history retained	Procedure/query
T-15	Build	Fresh exact-tag build	All manifest scripts and validation succeed	Build log

The repository contains expected-result SQL; actual results/pass-fail should be captured when MySQL is available and during the user's demonstration.

14. Requirement traceability

FR	Tables/relationships	Enforcement/objects	Report/test evidence
FR-01	BookTitle -> BookCopy -> Library; DirectorLibrary	sp_RegisterBookCopy, FK, scoped function	T-01, T-02, RQ-06
FR-02	Library -> BookCopy -> BookTitle -> Category	vw_AvailableCatalog, copy index	T-03, RQ-01
FR-03	Patron -> Loan -> BookCopy/Library	sp_CheckoutCopy, lock, transaction, active-copy UQ	T-04, T-05, RQ-02
FR-04	Loan -> return Library; BookCopy current Library	sp_ReturnCopy, transaction	T-06, T-07, RQ-03
FR-05	Donor -> Donation -> Item -> Title/Copy	donation procedures and checks	T-08, RQ-07
FR-06	Patron -> Visit -> Library/actor	sp_RecordVisit	T-09, RQ-05
FR-07	Patron -> Loan/Visit; DirectorLibrary	protected views, roles, scoped design	T-10, RQ-03/RQ-05
FR-08	Loan -> Copy -> Title/Library	date-range COUNT DISTINCT query	T-11, RQ-04
FR-09	Copy -> InventoryAdjustment -> Director	sp_AdjustInventory, immutable triggers	T-12, RQ-06/RQ-08
FR-10	All transactional relationships	functions, procedures, checks, rollback	T-02, T-05, T-07, T-10

FR	Tables/relationships	Enforcement/objects	Report/test evidence
FR-11	Actor -> AuditEvent -> affected record	procedure audit inserts, immutable triggers	T-13, RQ-08
FR-12	Library status and historical FKs	sp_SetLibraryStatus, procedure checks	T-14
FR-13	Loan and generated active-copy key	state validation, UNIQUE, locking read	T-05, T-07

Submission quality checklist

- FRD, TDD, ERD, schema, routines, tests, and sample data use the same entity and requirement names.
- The draw.io ERD contains every base entity, primary key, foreign key, and major relationship.
- Every table and column in the executable schema is described in this TDD.
- Every significant functional requirement maps to enforcement and test evidence.
- The manifest lists every SQL file once in dependency order.
- The build uses the immutable requested Git tag and records the resolved commit.
- The build refuses destructive replacement unless the operator explicitly opts in.
- All sample identities are fictional and all credentials remain outside the repository.
- Validation includes objects, row counts, reports, an execution plan, positive tests, and negative tests.
- The live demonstration remains a separate student activity, as requested.